

1.2 Measurements & Errors

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Hard

Time allowed: 60

Score: /42

Percentage: /100

Question 1a

In an experiment to determine the emissivity of a circular surface, the diameter d of the object is measured using the vernier scale shown in Fig. 1.1.

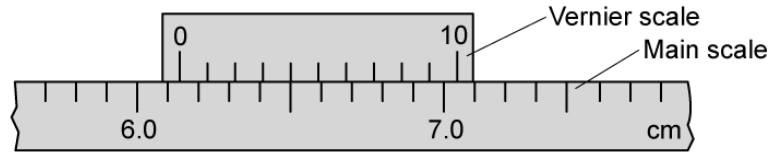


Fig. 1.1

Determine the diameter of the circular object from the reading in Fig. 1.1 and its associated uncertainty.

[2 marks]

Question 1b

The energy per unit time P radiated by an object with surface area A at absolute temperature T is given by

$$P = e\sigma AT^4$$

where e is the emissivity of the object and σ is the Stefan-Boltzmann constant.

Show that the emissivity e has no unit.

[3 marks]

Question 1c

As well as diameter d , the following measurements were taken:

$$P = 3.0 \pm 0.1 \text{ W}$$

$$T = 500 \pm 1 \text{ K}$$

Determine the value of the emissivity e of the surface and its uncertainty. Give your answer to an appropriate degree of precision.

[4 marks]

Question 2a

Fig. 1.1 shows a circuit designed by a student investigating the variation of potential difference with time as a capacitor charges and discharges whilst maintaining a constant current.

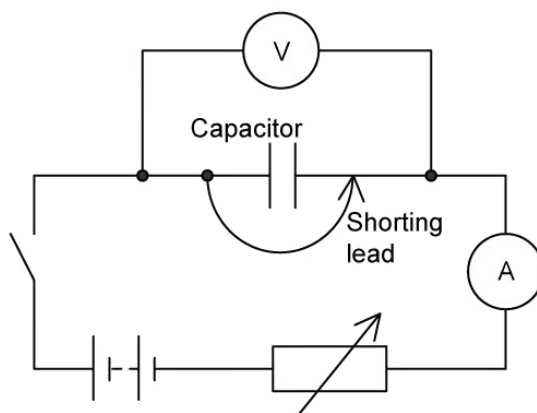


Fig. 1.1

The power supply has an e.m.f. of 6.0 V and negligible internal resistance. During the charging process, the maximum value of the resistance of the variable resistor is 100 k Ω .

- (i)
Suggest **one** advantage of using an analogue ammeter rather than a digital ammeter for this experiment.

[1]

- (ii)
Fig. 1.2 shows three analogue ammeters which can detect currents up to a maximum of 50 μ A, 100 μ A or 500 μ A.

Suggest, with a calculation, which of the analogue ammeters in Fig. 1.2 would be most suitable for the experiment.



Fig. 1.2

[3]

- (iii)
State the uncertainty in a reading of current using the ammeter you choose in (ii).

[1]

[5 marks]

Question 2b

A digital voltmeter with a three-digit display is used to measure the potential difference across the capacitor. A reading made during the charging process is shown in Fig. 1.3.



Fig. 1.3

The manufacturers of the meter state that its accuracy is $\pm 2\%$ and ± 1 digit.

- (i)
Determine the maximum possible value of the potential difference across the capacitor at the moment the reading in Fig. 1.3 was taken.
- (ii)
The reading on the voltmeter has high precision. State and explain why the reading may not be accurate.

[2]

[2]

[4 marks]

Question 2c

Initially, the shorting lead is connected across the capacitor, as shown in Fig. 1.1, to fully discharge it. The timer is started at the moment the shorting lead is disconnected as the capacitor begins to charge.

The student measures the potential difference across the capacitor at 5 s intervals while adjusting the variable resistor to keep the charging current constant.

The label on the capacitor states its capacitance C is $510\ \mu\text{F}$.

Determine the number of different readings the student will be able to take before the capacitor becomes fully charged.

The charge on the capacitor is equal to $Q = CV$

[3 marks]

Question 2d

The capacitor used in the experiment has a manufacturing tolerance of $\pm 10\%$. Throughout the experiment, the charging current is maintained at the value calculated in (a).

The data from the experiment produces a straight-line graph for the variation of potential difference with time. This shows that the potential difference across the capacitor increases at a rate of $110\ \text{mV s}^{-1}$.

(i)

Calculate the true capacitance of the capacitor.

[2]

(ii)

Deduce whether the capacitor is within the manufacturer's tolerance.

[2]

[4 marks]

Question 3a

An object of mass 50.00 g is placed on four different balances. For each balance, the reading is taken five times, as shown in Table 1.1.

Table 1.1

Balance	Mass Reading / g				
	1st	2nd	3rd	4th	5th
1	50.00	50.00	50.02	50.01	50.02
2	50.12	49.94	50.04	49.89	49.95
3	50.52	50.53	50.52	50.54	50.54
4	48.9	49.6	50.2	50.9	51.1

Analyse the data and conclude which sets of readings are accurate, precise or neither.

[2 marks]

Question 3b

A titanium sphere and a titanium cylinder were used in an experiment to measure the density of titanium. The diameter of the sphere was measured using a micrometer, and the diameter of the cylinder was measured using vernier callipers, as shown in Fig. 1.1.

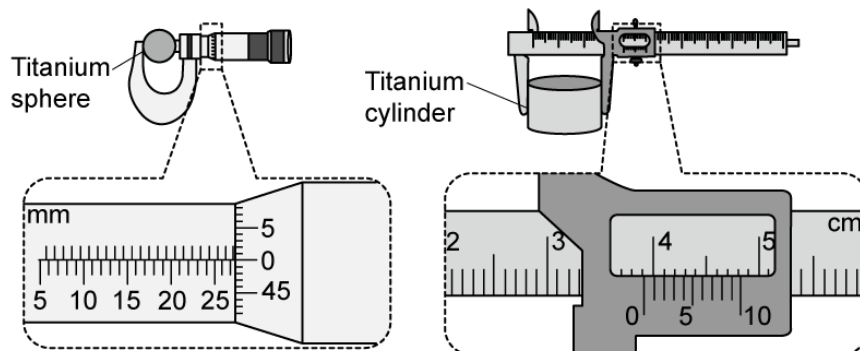


Fig. 1.1

Using Fig. 1.1, determine

(i)
the diameter of the titanium sphere, in cm, with its uncertainty.

[2]

(ii)
the diameter of the titanium cylinder, in cm, with its uncertainty.

[2]

[4 marks]

Question 3c

The measurements for the mass and height of the cylinder with their fractional uncertainties are shown in Table 1.2.

Table 1.2

	measurement	fractional uncertainty
cylinder mass / g	774.8	0.03
cylinder height / cm	15.0	0.01

The object used in **(a)** was the titanium sphere. The mass of the sphere was measured using the balance 2 shown in Table 1.1.

Determine

- (i)
the density of the titanium sphere, in g cm^{-3} , and its associated percentage uncertainty.

[4]

- (ii)
the density of the titanium cylinder, in g cm^{-3} , and its associated percentage uncertainty.

[3]

[7 marks]

Question 3d

The true value of the density of the titanium used in both the cylinder and the sphere was found to be 4.51 g cm^{-3} .

The student assumes that the percentage difference between the density of the titanium cylinder and the true value is solely due to the uncertainty in the volume and the uncertainty in the mass.

Assess the validity of this assumption.

[4 marks]

Model Answers: Hard

1a

Determine the diameter of the circular object and its associated uncertainty:

- Main scale reading = 6.1 cm
- Vernier scale reading = 0.04 cm

Calculate the final reading by adding the two readings:

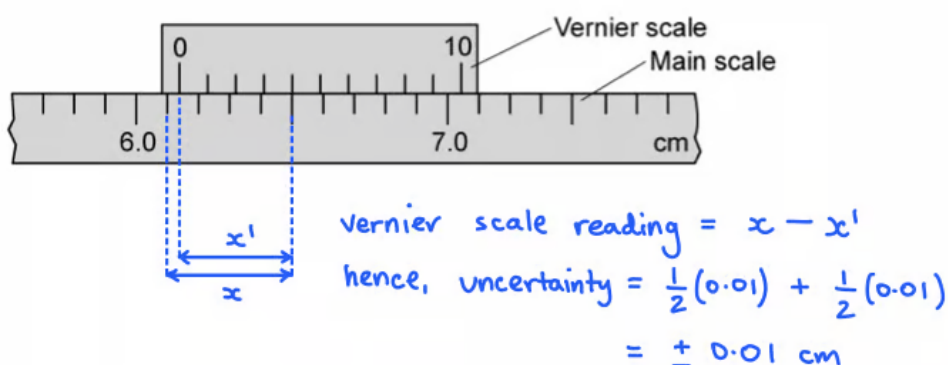
- Diameter, $d = 6.1 + 0.04 = 6.14$ cm [1 mark]

Determine the uncertainty in the diameter:

- Uncertainty in the measurement = $\pm$ smallest division
- The smallest division on the scale is 0.01 cm
- Hence, uncertainty in $d = \pm 0.01$ cm [1 mark]

[Total: 2 marks]

Measurements on instruments such as a rulers or vernier scales involve a subtraction between two points on the scale. The vernier part of the reading is really a subtraction, so we use $\pm$ (smallest division) rather than $\pm (\frac{1}{2} \times \text{smallest division})$



That means 0.01 cm is the smallest scale reading that can be made without estimation, hence the reading above is 6.14 ± 0.01 cm

1b

Show that the emissivity e has no unit:

- Units on RHS:
 - $[P] = W$ [1 mark]

- Units on LHS:

All correct for **one mark**

- Units of $[T]^4 = K^4$
- Units of $[A] = m^2$
- Units of $[\sigma] = W m^{-2} K^{-4}$

Compare the units on both sides:

- $[P] = [e][\sigma][A][T^4]$
- $W = (W m^{-2} K^{-4})(m^2)(K^4)$ [1 mark]
- RHS = LHS, hence e is unitless

[Total: 3 marks]

1c

Determine the value of the emissivity e of the surface and its uncertainty:

List the known quantities:

- Power, $P = 3.0 \pm 0.1 W$
- Temperature, $T = 500 \pm 1 K$
- Diameter, $d = 6.14 \pm 0.01 \text{ cm} = 0.0614 \pm 0.0001 \text{ m}$ (from **(a)**)
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} W m^{-2} K^{-4}$

Determine the value of emissivity, e :

- Cross-sectional area of a circle: $A = \left(\frac{\pi d^2}{4} \right)$

- $P = e\sigma AT^4 \Rightarrow P = e\sigma\left(\frac{\pi d^2}{4}\right)T^4 \Rightarrow e = \frac{4P}{\pi\sigma d^2 T^4}$
- Substitute: $e = \frac{4 \times 3.0}{\pi \times (5.67 \times 10^{-8}) \times (0.0614^2) \times (500^4)}$
- $e = 0.2859 = 0.29$ (2 s.f.) [1 mark]

Determine the uncertainty in emissivity, Δe :

- Separate the constants and the quantities: $e = \frac{4}{\pi\sigma} \left(\frac{P}{d^2 T^4} \right)$
- Propagate as fractional uncertainties: $\frac{\Delta e}{e} = \frac{\Delta P}{P} + 2 \frac{\Delta d}{d} + 4 \frac{\Delta T}{T}$
- $\frac{\Delta e}{e} = \frac{0.1}{3.0} + \frac{2 \times 0.01}{6.14} + \frac{4 \times 1}{500} = 0.04459$
- $\Delta e = 0.04459 \times e = 0.04459 \times 0.2859$
- $\Delta e = 0.01275 = 0.01$ (1 s.f.) [1 mark]

State the final answer to an appropriate degree of precision:

- $e = 0.29 \pm 0.01$ [1 mark]

[Total: 4 marks]

To propagate fractional uncertainties with and without quantities raised to a power:

$$\text{If } y = \frac{ab}{c} \text{ then } \frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$$

$$\text{If } y = a^n b^m \text{ then } \frac{\Delta y}{y} = \left(n \frac{\Delta a}{a} \right) + \left(m \frac{\Delta b}{b} \right)$$

This is why the fractional uncertainty in d^2 is $2 \times \frac{\Delta d}{d}$ and the fractional uncertainty in T^4 is $4 \times \frac{\Delta T}{T}$

When quoting values with their uncertainties, an 'appropriate degree of precision' means the uncertainty should not have a greater level of precision than the value

If e was quoted to 3 s.f., an appropriate degree of precision would be $e = 0.286 \pm 0.013$

2a

(i) An advantage of using an analogue ammeter is:

- The position on scale can be marked (as it needs to be kept constant) / easier to read quickly / easier to detect changes; [1 mark]

(ii) The most suitable ammeter for this experiment is the:

- 100 μA ammeter; [1 mark]

Calculation:

- Maximum resistance, $R = 100 \text{ k}\Omega = 100 \times 10^3 \Omega$
- e.m.f. of the supply, $V = 6.0 \text{ V}$
- Initial current: $V = IR \Rightarrow I = \frac{V}{R}$
- Initial current: $I = \frac{6.0}{100 \times 10^3} = 6 \times 10^{-5} \text{ A} = 60 \mu\text{A}$ [1 mark]

Conclusion:

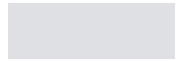
Any **one** from:

- 100 μA is high enough to detect the 60 μA current and is a suitable for detecting changes; [1 mark]
- 50 μA is too low to detect the 60 μA current; [1 mark]
- 500 μA might not be sensitive enough to detect readings / changes in readings; [1 mark]

(iii) Uncertainty in 100 μA ammeter readings:

- Smallest division = 2 μA

- Uncertainty in a reading = $\pm \frac{1}{2} \times \text{smallest division} = \pm (\frac{1}{2} \times 2)$
- Uncertainty in a reading = $\pm 1 \mu\text{A}$ [1 mark]



(iii) Uncertainty in 100 μA ammeter readings:

- Smallest division = 2 μA
- Uncertainty in a reading = $\pm \frac{1}{2} \times \text{smallest division} = \pm(\frac{1}{2} \times 2)$
- Uncertainty in a reading = $\pm 1 \mu\text{A}$ [1 mark]

[Total: 5 marks]

2b

(i) Determine the maximum possible value of V :

- Absolute uncertainty in $V = 2\%$ of 2.23 V
- Absolute uncertainty in $V = \frac{2}{100} \times 2.23 = \pm 0.04 \text{ V}$ [1 mark]
- Max. value of $V = 2.23 + 0.04 + 0.01 = 2.28 \text{ V}$ [1 mark]

(ii) The reading may not be accurate because:

- There may be a zero error / calibration error / systematic error; [1 mark]
- Which makes all readings either higher or lower than true value; [1 mark]

[Total: 4 marks]

Don't forget to add the ± 0.01 in part (i)!

Where the question says the meter has an accuracy of ± 1 digit, this refers to the uncertainty in the final digit of the three-digit display on top of the uncertainty of 2%, hence the total uncertainty is $0.04 + 0.01 = \pm 0.05 \text{ V}$

2c

Determine the number of different readings the student will be able to take:

List the known quantities:

- Capacitance, $C = 510 \mu\text{F} = 510 \times 10^{-6} \text{ F}$
- Potential difference when fully charged, $V = 6.0 \text{ V}$
- Current, $I = 60 \mu\text{A} = 60 \times 10^{-6} \text{ A}$

Determine the total charge on the capacitor when fully charged:

- Total charge: $Q = CV = (510 \times 10^{-6}) \times 6.0$
- $Q = 3.06 \times 10^{-3} \text{ C} = 3.06 \text{ mC}$ [1 mark]

Determine the time taken for the capacitor to fully charge:

- Current: $I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I}$
- Time: $t = \frac{3.06 \times 10^{-3}}{60 \times 10^{-6}} = 51 \text{ s}$ [1 mark]

If a reading is taken every 5 s...

- Number of readings = $\frac{51}{5} = 10.2$
- Number of readings = 10 [1 mark]

[Total: 3 marks]

2d

List the known quantities:

- Charging current, $I = 60 \mu\text{A} = 60 \times 10^{-6} \text{ A}$
- Rate of p.d. increase, $\frac{V}{t} = 110 \text{ mV s}^{-1} = 110 \times 10^{-3} \text{ V s}^{-1}$
- Tolerance / uncertainty in capacitance = $\pm 10\%$
- Quoted value = $510 \mu\text{F}$

(i) Calculate the true capacitance of the capacitor:

- Total charge: $I = \frac{Q}{t} \Rightarrow Q = It = (60 \times 10^{-6}) \times t$
- Rate of p.d. increase = $\frac{V}{t} \Rightarrow V = (110 \times 10^{-3}) \times t$

- Capacitance: $Q = CV \Rightarrow C = \frac{Q}{V}$
- $C = \frac{(60 \times 10^{-6}) \times \cancel{f}}{(110 \times 10^{-3}) \times \cancel{f}}$
- $C = 5.45 \times 10^{-4} \text{ F} = 545 \mu\text{F}$

(ii) Deduce whether the capacitor is within the manufacturer's tolerance:

EITHER

Method 1: Determine the uncertainty in C

- Absolute uncertainty in $C = 510 \times \frac{10}{100} = \pm 51 \mu\text{F}$ [1 mark]
- The capacitor is within the manufacturer's tolerance because $545 \mu\text{F}$ lies within the range $459 - 561 \mu\text{F}$; [1 mark]

OR

Method 2: Determine the percentage difference

- Percentage difference = $\frac{545 - 510}{510} \times 100\% = 6.9\%$; [1 mark]
- The % difference between the values of capacitance is 6.9% which is within 10% ; [1 mark]

[Total: 4 marks]

3a

Analysis of the data:

- Take a mean of each set of readings; [1 mark]
- Write a conclusion about each set of readings using the words accurate, precise, inaccurate and imprecise with correct reasoning; [1 mark]

Example of analysis:

- Balance 1 mean = 50.01 g
 - Balance 2 mean = 49.99 g
 - Balance 3 mean = 50.53 g
 - Balance 4 mean = 50.1 g
-
- Balances 1 & 2 show accurate readings
 - Because their means are close to the true value
-
- Balances 1 & 3 show precise readings
 - Because all readings are close to their mean value
-
- Balances 3 & 4 show inaccurate readings
 - Because their mean values are not as close to the true value as the 1 & 2
-
- Balances 2 & 4 show imprecise readings
 - Because the readings are widely scattered about their mean value
 - Because balance 4 measures to a lower level of precision / number of decimal places than the other three

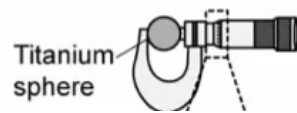
[Total: 2 marks]

3b

(i) Determine the diameter of the titanium sphere with its uncertainty:

- From Fig. 1.1:
Reading on the barrel = 27.5 mm
Reading on the thimble = 0
Final reading = 27.50 mm
- Diameter of the sphere = 2.750 cm **[1 mark]**
- Uncertainty in the reading = $\pm$ smallest division
- Uncertainty in the reading = ± 0.01 mm = ± 0.001 cm **[1 mark]**

$\therefore$ Diameter of the titanium sphere = 2.750 ± 0.001 cm



- Balances 1 & 3 show precise readings
- Because all readings are close to their mean value
- Balances 3 & 4 show inaccurate readings
- Because their mean values are not as close to the true value as the 1 & 2
- Balances 2 & 4 show imprecise readings
- Because the readings are widely scattered about their mean value
- Because balance 4 measures to a lower level of precision / number of decimal places than the other three

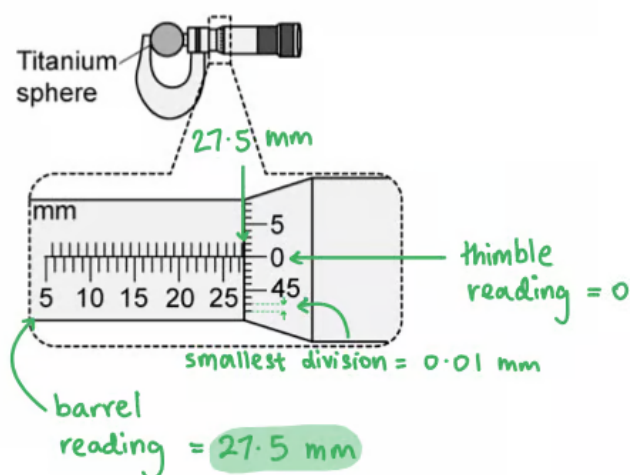
[Total: 2 marks]

3b

(i) Determine the diameter of the titanium sphere with its uncertainty:

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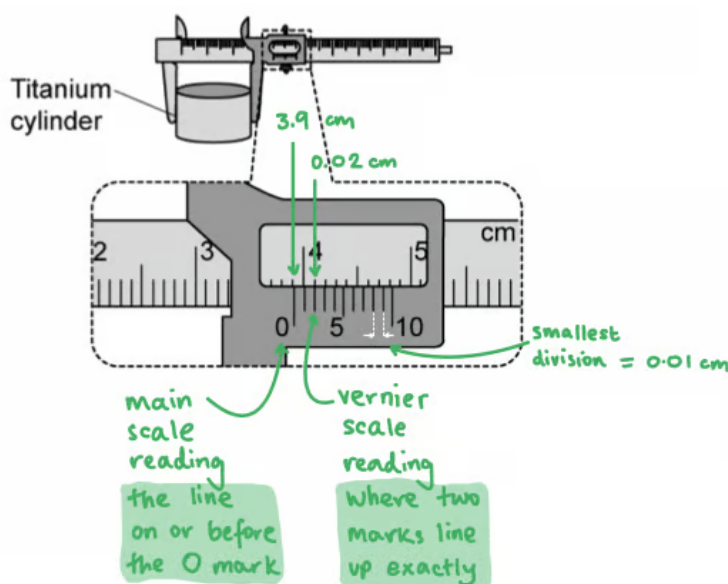
$\therefore$ Diameter of the titanium sphere = 2.750 ± 0.001 cm



(ii) Determine the diameter of the titanium cylinder with its uncertainty:

- From Fig. 1.1:
Reading on the main scale = 3.9 cm
Reading on the vernier scale = 0.02 cm
Final reading = 3.92 cm
- Diameter of the cylinder = 3.92 cm [1 mark]
- Uncertainty in the reading = $\pm$ smallest division
- Uncertainty in the reading = ± 0.01 cm [1 mark]

$\therefore$ Diameter of the titanium cylinder = 3.92 ± 0.01 cm



[Total: 4 marks]

3c

(i) Determine the density of the titanium sphere:

List the known quantities:

- Sphere diameter, $d = 2.750 \pm 0.001$ cm (from **(b)**)
- Mean sphere mass, $m = 49.99$ g (from **(a)**)

Calculate the density of the sphere:

- Volume of a sphere: $V = \frac{4}{3}\pi r^3 = \frac{4\pi}{3}\left(\frac{d}{2}\right)^3$
- Density: $\rho = \frac{m}{V} = \frac{3m}{4\pi}\left(\frac{2}{d}\right)^3$
- Density: $\rho = \frac{3 \times 49.99}{4\pi}\left(\frac{2}{2.750}\right)^3$ [1 mark]
- Density: $\rho = 4.5908 = 4.59 \text{ g cm}^{-3}$ [1 mark]

Convert the uncertainties into percentage uncertainties:

- Uncertainty in mass, $\Delta m = (\frac{1}{2} \times \text{range}) = \frac{1}{2} \times (50.12 - 49.89) = \pm 0.115 \text{ g}$ [1 mark]
- % uncertainty in m : $\% \Delta m = \frac{0.115}{49.99} \times 100\% = 0.23\%$
- % uncertainty in d : $\% \Delta d = \frac{0.001}{2.750} \times 100\% = 0.036\%$

Calculate the percentage uncertainty in density:

- $\% \Delta \rho = \% \Delta m + 3(\% \Delta d)$
- $\% \Delta \rho = 0.23 + (3 \times 0.036) = 0.34\%$ [1 mark]

Quote the density with its associated percentage uncertainty:

- Density of titanium sphere, $\rho = 4.59 \text{ g cm}^{-3} \pm 0.34\%$

(ii) Determine the density of the titanium cylinder:

List the known quantities:

- Cylinder diameter, $d = 3.92 \pm 0.01 \text{ cm}$ (from (b))
- Cylinder mass, $m = 774.8 \text{ g}$
- Cylinder height, $h = 15.0 \text{ cm}$
- Fractional uncertainty in mass, $\Delta m/m = 0.03$
- Fractional uncertainty in height, $\Delta h/h = 0.01$

Calculate the density of the cylinder:

- Volume of a cylinder: $V = \pi r^2 h = \pi h \left(\frac{d}{2} \right)^2$
- Density: $\rho = \frac{m}{V} = \frac{m}{\pi h \left(\frac{d}{2} \right)^2}$
- Density: $\rho = \frac{774.8}{\pi \times 15.0 \left(\frac{2}{3.92} \right)^2}$ [1 mark]
- Density: $\rho = 4.2799 = 4.28 \text{ g cm}^{-3}$ [1 mark]

Calculate the percentage uncertainty in density:

- $\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta h}{h} + 2 \left(\frac{\Delta d}{d} \right)$
- $\frac{\Delta \rho}{\rho} = 0.03 + 0.01 + 2 \left(\frac{0.01}{3.92} \right) = 0.045$
- $\% \Delta \rho = 0.045 \times 100\% = 4.5\%$ [1 mark]

Quote the density with its associated percentage uncertainty:

- Density of titanium cylinder = $4.28 \text{ g cm}^{-3} \pm 4.5\%$

[Total: 7 marks]

Remember, if a quantity has a power associated with it then you must multiply the propagated uncertainty by that power. This is because a quantity with a power has a larger effect on the calculated quantity, hence its fractional uncertainty will be higher.

In the case of the cylinder, the diameter is squared, hence its fractional, or percentage, uncertainty is twice as large since the measured diameter has such a great effect on the final value of ρ .

In the case of the sphere, the diameter is cubed which has an even larger effect on the final value of ρ as the fractional uncertainty must be multiplied by 3.

3d

- Volume of a cylinder: $V = \pi r^2 h = \pi h \left(\frac{d}{2} \right)^2$
- Density: $\rho = \frac{m}{V} = \frac{m}{\pi h \left(\frac{d}{2} \right)^2}$
- Density: $\rho = \frac{774.8}{\pi \times 15.0 \left(\frac{2}{3.92} \right)^2}$ [1 mark]
- Density: $\rho = 4.2799 = 4.28 \text{ g cm}^{-3}$ [1 mark]

Calculate the percentage uncertainty in density:

- $\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta h}{h} + 2 \left(\frac{\Delta d}{d} \right)$
- $\frac{\Delta \rho}{\rho} = 0.03 + 0.01 + 2 \left(\frac{0.01}{3.92} \right) = 0.045$
- $\% \Delta \rho = 0.045 \times 100\% = 4.5\%$ [1 mark]

Quote the density with its associated percentage uncertainty:

- Density of titanium cylinder = $4.28 \text{ g cm}^{-3} \pm 4.5\%$

[Total: 7 marks]

Remember, if a quantity has a power associated with it then you must multiply the propagated uncertainty by that power. This is because a quantity with a power has a larger effect on the calculated quantity, hence its fractional uncertainty will be higher.

In the case of the cylinder, the diameter is squared, hence its fractional, or percentage, uncertainty is twice as large since the measured diameter has such a great effect on the final value of ρ .

In the case of the sphere, the diameter is cubed which has an even larger effect on the final value of ρ as the fractional uncertainty must be multiplied by 3.

3d

List the known quantities:

- True value of density = 4.51 g cm^{-3}
- Sphere density = 4.59 g cm^{-3}
- Cylinder density = 4.28 g cm^{-3}
- % uncertainty in sphere density = 0.3%
- % uncertainty in cylinder density = 4.5%

Calculate the percentage difference from the sphere's density and the true value:

- % difference (sphere) = $\frac{4.59 - 4.51}{4.51} \times 100\% = 1.8\%$

Calculate the percentage difference from the cylinder's density and the true value:

- % difference (cylinder) = $\frac{4.51 - 4.28}{4.51} \times 100\% = 5.1\%$

Compare the percentage difference with the calculated uncertainty for the sphere:

- Sphere: $1.8 - 0.3 = 1.5\%$ [1 mark]
- There is a significant error ($>1\%$) unaccounted for by the mass or volume, hence the assumption may not be valid for the sphere; [1 mark]

Compare the percentage difference with the calculated uncertainty for the cylinder:

- Cylinder: $5.1 - 4.5 = 0.6\%$ [1 mark]
- There is only a small error ($<1\%$) unaccounted for by the mass or volume, hence the assumption can be said to be valid for the cylinder; [1 mark]

[Total: 4 marks]

Always read the question carefully, in this case, you are trying to find out if the student's assumption is correct.

According to the density equation ($\rho = m/V$), ρ **only** depends on m and V , so the percentage difference and the percentage uncertainties calculated in (c) should be the same if the assumption is correct. In other words, the difference between the percentage difference with the true value and the uncertainties calculated in (c) should be close to zero for the assumption to be valid.

You may have automatically assumed that this assumption must be correct, but working through the maths shows it isn't. Measurement is a science in itself, and there are many ways that errors and uncertainties creep in, such as random errors or

unknowns such as impurities in the metal

